

KINGSFORD, MI, USA

WMO: 727437

Lat: 45.818N

Lon: 88.114W

Elev: 342

StdP: 97.28

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.6	-28.2	0.3	-23.2	-26.0	0.4	-20.8	11.0	-7.9	9.6	-8.6	1.7	280	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	31.1	21.5	29.2	20.2	27.7	19.4	23.1	28.2	21.9	26.9	20.8	25.5	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.7	25.8	20.3	15.6	24.6	19.0	14.4	23.6	70.2	28.2	65.7	26.8	61.9	25.9	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.4	-28.5	33.9	2.5	1.6	-30.3	35.0	-31.8	36.0	-33.2	36.9	-35.0	38.1		
(5)			-28.7	24.9	2.4	1.3	-30.4	25.8	-31.9	26.6	-33.2	27.3	-35.0	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-9.3	-8.0	-2.3	4.8	11.9	17.3	19.9	18.6	14.5	7.3	0.3	-6.2
	DBStd	11.47	6.29	6.15	6.40	5.04	4.88	4.06	3.51	3.34	4.50	4.82	5.12	5.97
	HDD10.0	2613	598	504	384	169	36	2	0	0	11	113	293	504
	HDD18.3	4755	856	737	640	407	209	67	25	38	130	342	541	762
	CDD10.0	1078	0	0	3	12	93	219	306	266	147	31	2	0
	CDD18.3	177	0	0	0	0	8	34	72	45	16	1	0	0
	CDH23.3	2059	0	0	3	12	149	428	817	465	171	14	0	0
	CDH26.7	583	0	0	0	2	41	124	263	113	38	2	0	0
(14) Wind	WSAvg	3.0	3.2	3.3	3.2	3.4	3.3	2.8	2.7	2.5	2.8	3.1	3.2	3.1
(15) Precipitation	PrecAvg	710	31	24	39	55	78	85	82	85	86	66	46	35
	PrecMax	939	102	63	104	113	170	164	288	174	155	159	138	89
	PrecMin	476	2	1	6	6	10	25	24	6	18	10	8	5
	PrecStd	115	21	17	26	25	37	34	44	34	37	36	29	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	9.8	20.8	25.1	30.6	32.7	33.7	32.1	30.7	25.5	17.7	9.0
		MCWB	2.5	4.8	14.1	14.6	18.9	22.6	22.9	22.2	21.2	17.3	11.6	5.7
	2%	DB	3.0	5.5	13.8	21.0	27.4	29.9	31.3	29.3	27.4	21.7	12.9	5.1
		MCWB	1.1	2.6	8.7	12.1	17.0	20.3	21.8	20.6	19.5	15.0	9.3	2.9
	5%	DB	1.3	2.7	10.0	17.6	24.3	27.8	29.5	27.6	25.3	18.5	10.2	2.8
		MCWB	0.1	0.6	5.4	9.5	15.3	19.4	20.7	19.6	18.1	14.3	7.1	1.5
	10%	DB	-0.1	1.0	7.1	14.7	21.6	26.0	27.7	26.1	22.8	15.7	7.9	1.4
		MCWB	-1.1	-0.5	3.5	7.7	13.9	17.8	19.8	18.6	17.2	11.9	5.5	0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	5.8	14.3	15.9	21.0	23.9	25.2	24.1	22.6	19.9	12.8	6.5
		MCDB	4.8	9.2	18.1	22.2	26.9	29.4	30.6	28.8	27.3	22.7	15.9	7.4
	2%	WB	1.4	2.8	9.6	13.0	18.9	22.0	23.4	22.4	20.9	16.6	10.2	3.3
		MCDB	2.8	5.0	13.1	19.2	24.5	27.6	28.7	27.0	25.3	20.3	12.8	4.6
	5%	WB	0.2	0.8	6.1	10.3	17.1	20.5	22.2	21.3	19.7	14.4	7.6	1.7
		MCDB	1.3	2.4	9.4	15.6	21.8	25.3	27.1	25.6	23.2	17.5	9.4	2.7
	10%	WB	-1.1	-0.6	3.6	8.4	15.1	19.1	20.9	20.1	18.3	12.2	5.6	0.4
		MCDB	-0.1	0.9	6.6	13.9	20.4	23.7	26.0	24.2	21.6	15.1	7.9	1.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.5	11.3	12.9	13.9	13.2	13.5	12.9	12.7	10.7	8.4	8.0
		MCDBR	9.2	11.4	14.8	19.4	17.8	15.3	15.0	14.6	15.4	15.2	12.7	9.3
	5% WB	MCWBR	7.8	8.8	9.8	11.3	9.0	7.4	7.0	7.0	8.1	9.4	9.0	7.6
		MCWBR	8.8	10.0	13.4	17.2	14.8	13.3	13.3	12.0	12.2	13.2	11.2	8.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.242	0.275	0.313	0.338	0.373	0.371	0.371	0.372	0.350	0.314	0.286	0.254
		taud	2.425	2.317	2.360	2.405	2.355	2.383	2.403	2.393	2.462	2.557	2.540	2.462
	Ebn at Noon	Ebn at Noon	890	909	918	921	894	895	890	876	869	860	828	834
		Edh at Noon	68	93	104	110	120	117	113	110	94	74	62	60
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.38	2.39	3.66	4.57	5.46	5.89	6.07	5.13	3.93	2.40	1.38	1.01
	RadStd	RadStd	0.13	0.27	0.35	0.45	0.44	0.47	0.29	0.33	0.26	0.26	0.15	0.10

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (39 neighbors)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page